SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.4: Transport in Plants — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.4: Transport in Plants
Driving Question	DRIVING QUESTION: How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon: The Travelling Red Dye & The Sweating Bag**	Students observe two phenomena: (1) a white-flowered plant (or celery stalk) placed in red-dyed water develops distinct red streaks only in specific channels of the stem and petals — not uniformly throughout the tissue; (2) a sealed, clear polythene bag placed over a leafy maize shoot accumulates visible water droplets on its inner surface within 20–30 minutes, even in a dry classroom. Teachers should cold-call at least 4 students to describe exactly where the red colour appears and where the water droplets come from before moving on.	Students learn that water does NOT move randomly through the entire stem — it travels through specific, distinct channels (later identified as xylem vessels). They also learn that living leaves continuously release water vapour into the surrounding air — a process they will later name transpiration. Both observations anchor the unit's driving question: something is actively moving water from soil to leaf and then releasing it.	The red dye is carried exclusively through xylem vessels — narrow, continuous tubes that run from root to stem to leaf veins. Because xylem is the only pathway for upward water movement, the dye colours only those channels. The water droplets inside the polythene bag come from water vapour escaping through tiny pores (stomata) on the leaf surface; the bag traps the vapour, which then condenses. Pedagogical note: Do NOT fully name or explain osmosis, xylem, or transpiration here — preserve those revelations for Lessons 2–5. Redirect student over-explanations back to the DQB as open questions.
Lesson 2 **Structures for Absorption: Root Hair Cells, Xylem & Phloem, and Capillary Forces**	Students examine prepared slides or high-quality diagrams of (1) a longitudinal section of a maize root tip showing root hair cells projecting into soil spaces, and (2) a cross-section of a maize stem showing the distinct vascular bundles containing xylem and phloem. Teachers direct students to measure (with a ruler on the diagram) the relative length of a root hair cell versus a	Students learn that root hair cells have a greatly extended surface area (the hair projection increases contact with soil water by up to 600%) that maximises water absorption from the soil. Xylem vessels are narrow, hollow, dead cells with lignified walls arranged end-to-end, forming continuous tubes ideal for water transport. Phloem cells are living and transport	The elongated shape of root hair cells is a structural adaptation: greater surface area means more membrane available for water and ion uptake per unit of root length. Xylem vessels are dead at maturity because their end walls break down, creating an unobstructed hollow column; lignin reinforcement prevents the tube from collapsing under tension. The capillary

	normal epidermal cell, and to count the number of vascular bundles visible in the cross-section. A capillary rise demonstration — thin glass tubes of varying diameters placed in coloured water — is observed alongside, with students recording the height water reaches in each tube.	dissolved sugars. Capillary action — the tendency of water to rise in narrow tubes due to adhesion and cohesion — is one physical force that assists water movement up the xylem.	demonstration explains why narrower xylem vessels (as in small herbaceous plants) can draw water higher — the same principle used in the thin wicks of a kerosene lamp common in rural Kenyan homes. Pedagogical note: Explicitly connect the xylem channels seen here back to the red-dye streaks from Lesson 1 — ask: 'Which structure did the dye travel through?' Allow 30 seconds wait time before cold-calling.
Lesson 3 **Absorption of Water and Mineral Salts: Osmosis, Active Transport, and Root Pressure**	Students conduct or observe two demonstrations: (1) Osmosis — a raw potato or maize grain cylinder is placed in dilute salt water vs. plain water; students record mass/length changes after 20 minutes, noting that the cylinder in plain water swells and the one in salt water shrinks. (2) Root pressure — a freshly cut maize stem stump (or a potted plant with the shoot removed) is observed to slowly exude water droplets from the cut surface over several minutes, even without any pulling force from leaves above. Teachers should ask students to predict outcomes BEFORE each observation, recording predictions on the DQB.	Students learn that water enters root hair cells by OSMOSIS — it moves from a region of higher water potential (dilute soil water) to a region of lower water potential (concentrated cell sap) across a semi-permeable membrane, requiring NO energy. Mineral salts (e.g., nitrates and phosphates vital for maize growing in Rift Valley soils) are absorbed by ACTIVE TRANSPORT — a process that requires ATP energy from the cell and moves ions against their concentration gradient. Root pressure is the positive pressure generated in the xylem by continuous osmotic water entry at the roots, which can push water upward even in the absence of transpiration.	Osmosis is a special case of diffusion involving water molecules only, across a selectively permeable membrane, down a water potential gradient. The potato/maize cylinder results mirror what happens in root hair cells: when soil water is more dilute than cell sap, water enters by osmosis, creating turgor pressure. Active transport explains how mineral ions that are already more concentrated inside the root than in soil water can still be absorbed — the cell expends metabolic energy (from cellular respiration) to pump them in via carrier proteins. Root pressure, while insufficient alone to push water to the top of a tall maize plant, contributes to the overall driving force. Pedagogical note: Reinforce the distinction between osmosis (passive, no energy) and active transport (energy-dependent) using a local analogy — water flowing downhill (osmosis) vs. a porter carrying goods uphill to Kericho market (active transport).
Lesson 4 **Translocation: How Does the Maize Plant Move Sugar from Leaf to Root?**	Students observe a ring-barking (girdling) demonstration or photograph: a maize or tree stem from which a complete ring of bark (phloem-containing) has been removed. Students note that after several days, a swelling of tissue appears immediately ABOVE the cut ring but not below it. A second observation uses a sugar solution placed on a leaf surface — a few hours later, aphids (or a simulated aphid feeding model) are found clustered on the phloem of that stem section, confirming sugar is moving in the phloem.	Students learn that plants have TWO distinct transport systems: xylem (upward, water and minerals, dead cells, driven by transpiration pull and root pressure) and phloem (bidirectional but primarily downward from leaf to root, dissolved sugars and other organic compounds, living cells). The movement of sugars and other assimilates through phloem is called TRANSLOCATION. The ring-barking result proves that sugar cannot pass the removed bark (phloem) and accumulates above — confirming phloem as the sugar highway.	The pressure-flow (mass flow) hypothesis explains translocation: sugar loaded into phloem at the source leaf by active transport raises osmotic concentration, drawing water in from adjacent xylem, building up pressure; at the sink, sugar is unloaded (used for respiration or stored as starch in the maize grain), reducing concentration and pressure; the resulting pressure gradient drives bulk flow from source to sink. This is why a developing maize cob — a powerful sink — can receive sugar from leaves metres away.

	Teachers cold-call 3 students: 'What does the swelling above the ring tell us about the direction sugar moves?'	Phloem transport is driven by a pressure-flow mechanism: high sugar concentration at the source (photosynthesising leaf) creates high osmotic pressure; low sugar concentration at the sink (growing root, maize cob) creates low pressure; water follows the sugar, generating bulk flow.	Pedagogical note: Use the analogy of water flowing through an irrigation channel from a high-pressure water tank (source) to a farm plot (sink) — familiar to students in Mwea or Perkerra irrigation schemes. Revisit Lesson 1's DQB: the phloem answers the question of how nutrients (not just water) reach all parts of the plant.
Lesson 5 **Transpiration Experiment Setup: Designing Investigations into Factors Affecting Water Loss in Plants**	Students set up a controlled transpiration investigation using potted maize seedlings or leafy shoots (e.g., from a local shrub). Variables tested across groups: (1) control — normal conditions; (2) Vaseline applied to lower leaf surface (blocking stomata); (3) Vaseline applied to upper leaf surface; (4) a fan providing gentle wind; (5) a plastic bag covering the shoot (high humidity). Mass of each pot+plant is recorded at setup (T=0) and again after 30–40 minutes. Students also use a potometer (or a simple syringe-and-ruler setup) to measure the rate of water uptake as a proxy for transpiration rate. Teachers ensure every group records a prediction BEFORE mass measurements.	Students learn that transpiration — the loss of water vapour from plant surfaces, primarily through stomata on the lower leaf surface — can be affected by multiple environmental factors: (1) blocking lower stomata (Vaseline on lower surface) dramatically reduces water loss compared to blocking the upper surface, proving most stomata are on the lower epidermis; (2) increased wind speed increases the transpiration rate by removing the humid boundary layer around the leaf; (3) high humidity (bag) reduces transpiration by decreasing the water vapour concentration gradient between the leaf interior and the air. Students learn to distinguish the independent variable, dependent variable, and controlled variables in their own experimental design.	Stomata are the primary route of water vapour exit — up to 90% of transpired water leaves through stomatal pores. The driving force is the water vapour concentration gradient: inside the leaf (sub-stomatal air spaces), the air is nearly 100% saturated with water vapour; outside, it is drier. When guard cells are turgid (daytime, in light), stomata open; when flaccid (night, drought), they close. Wind increases transpiration by continuously replacing the humid air just outside the stomata with drier air, steepening the gradient. High humidity reduces the gradient, slowing transpiration. Pedagogical note: Link to real Kenyan farming concerns — a maize farmer in Kitale during a dry spell sees wilting because transpiration exceeds water absorption; understanding these factors informs irrigation timing decisions.
Lesson 6 **Transpiration Results Analysis: Stomata, Guard Cells, and Environmental Factors**	Students collect and compare the mass-loss data recorded at the end of Lesson 5 across all experimental groups. They also examine microscope slides or printed micrographs of the lower epidermis of a maize leaf, identifying stomatal pores and the paired kidney-shaped guard cells surrounding each pore. Open vs. closed stomata are compared side by side. Teachers display a class data table on the board (or chalkboard) and cold-call 5 students to read out their group's results before the class calculates averages. Students graph transpiration rate (y-axis) against the treatment condition (x-axis) as a bar chart.	Students learn the detailed mechanism by which guard cells control stomatal opening and closing: when guard cells absorb water by osmosis (due to active uptake of potassium ions in light), they become turgid and their unevenly thickened walls cause them to bow outward, opening the pore; when they lose water (darkness, water stress), they become flaccid and the pore closes. Students also consolidate the relationship between four key environmental factors and transpiration rate — light intensity, temperature, humidity, and wind speed — using their own experimental data as evidence. They practise interpreting bar charts and identifying anomalous results in class data.	Guard cell mechanism is a feedback-regulated system: light triggers the H^+-ATPase pump in guard cells, which actively transports H^+ out and K^+ in; raised internal K^+ concentration lowers water potential inside the guard cell; water enters by osmosis, increasing turgor; the asymmetrically thickened inner wall (thicker on the side facing the pore) means that as the cell swells, it curves outward, widening the pore. This elegant mechanism allows the plant to balance CO_2 uptake for photosynthesis against water loss. Pedagogical note: Emphasise that this lesson closes the experimental loop opened in Lesson 5 — data collection and analysis are separate cognitive phases. Remind students that their graph is evidence they

			will use on the DQB to answer the driving question's 'what happens to that water in the end?' clause.
Lesson 7 **Importance of Transpiration and Community Application**	Students observe and discuss three local scenarios presented as photographs or short case studies: (1) a thermometer placed inside a dense maize crop in Kitale vs. one in an open, bare field on a hot afternoon — the crop reading is cooler; (2) satellite imagery or a map showing that areas around Mount Kenya forests and Aberdare ranges receive higher rainfall than deforested areas nearby (teacher narrates the water cycle connection); (3) a wilted maize plant at midday vs. the same plant in the evening — students observe that wilting is often reversible, suggesting it is a temporary transpiration–absorption imbalance. Teachers cold-call students: 'Name one way a Kenyan farmer could reduce harmful excess transpiration during a drought.'	Students learn that transpiration serves three critical functions for the maize plant: (1) COOLING — evaporation of water from leaf surfaces absorbs latent heat, keeping leaf temperatures 2–5°C below ambient air temperature on hot Kenyan afternoons, protecting photosynthetic enzymes; (2) CREATING TRANSPIRATION PULL — the continuous loss of water vapour at the leaf generates a negative pressure (tension) that pulls water up the xylem column from roots to leaves, acting as the main engine of upward water transport (cohesion-tension theory); (3) AIDING MINERAL TRANSPORT — the mass flow of water through xylem carries dissolved mineral salts from roots to all aerial parts of the plant. Students also learn the broader ecological importance: forests and crops release water vapour that contributes to local rainfall patterns and the regional water cycle.	The cohesion-tension theory explains how water reaches the topmost leaves of a tall maize plant against gravity: transpiration at the leaf surface creates a water deficit in mesophyll cells; water is drawn from xylem into mesophyll by osmosis; this creates tension (negative pressure) in the xylem; because water molecules are strongly cohesive (hydrogen bonding) and adhere to xylem walls, the entire water column is pulled up as a continuous thread — like pulling a chain upward by tugging its top link. This theory, formulated by Dixon and Joly, is supported by the observation that cutting a xylem vessel under tension causes an audible click as the column snaps. Pedagogical note: This lesson directly answers BOTH parts of the driving question — the mechanism of upward transport AND the ultimate fate of water (lost as vapour through stomata). Use the DQB to celebrate which questions have now been fully answered and which remain open for Lesson 8.
Lesson 8 **Synthesis and Assessment: Explaining the Complete Journey of Water in the Maize Plant**	Students revisit ALL physical evidence and data collected across Lessons 1–7: the red-dye stained stem cross-section, the polythene bag water droplets, the osmosis potato/maize cylinder results, the ring-barking photograph, the transpiration bar charts, and the guard cell micrographs. The teacher displays the original DQB from Lesson 1 and, working as a class, students vote on which questions have been fully answered, partially answered, or remain open. Each student then individually draws and annotates a complete diagram of a maize plant showing the full pathway of water — from soil, through root hair cell, across cortex, into xylem, up to leaf, through stomata, into air — labelling every process: osmosis, active transport, root pressure, capillary action, cohesion-tension,	Students consolidate the complete, integrated account of water and nutrient transport in a maize plant: water enters via osmosis into root hair cells → crosses the root cortex → enters xylem by osmosis and root pressure → travels up xylem driven primarily by transpiration pull (cohesion-tension) → reaches mesophyll cells in leaves → water vapour exits through stomata as transpiration. Mineral salts enter by active transport alongside water and are carried passively up the xylem. Sugars produced by photosynthesis in leaves are translocated downward through phloem by pressure flow to sinks such as roots, growing cobs, and seeds. Students also articulate the ecological and agricultural significance of these processes for Kenyan farming communities.	The driving question is now fully answered: a maize plant in a Kenyan farm moves water from soil to topmost leaves through a coordinated system of osmosis, active transport, root pressure, capillary action, and — most powerfully — the cohesion-tension generated by transpiration. Mineral nutrients hitchhike in the xylem water stream. The water's ultimate fate is evaporation as water vapour through open stomata — a process that simultaneously cools the leaf, drives the water column upward, and contributes water vapour to the local atmosphere above Kenyan farmlands. Pedagogical note: The annotated diagram serves as both a formative assessment tool and a cumulative model — teachers should check that every student's diagram includes all six labelled processes and connects

	and transpiration.		back to the original anchoring phenomena (red dye + sweating bag). Award credit for accurate use of Kenyan context (e.g., labelling the soil as 'Rift Valley red soil', the water source as 'irrigation from Mwea canal', or the atmosphere as 'humid air above Kitale maize belt').